

Python Backend Roadmap

- **Python Language Basics**

1. If statements
2. Comprehension
3. Lists
4. functions

- **Python in Advance Features**

1. object - oriented programming
2. Meta Class
3. Generators
4. Context Managers
5. Iterators

- **Python other features to learn**

1. Asynchronous Programming
2. Global Interpreter Lock (python)
3. Multiprocessing
4. Multithreading
5. Python Modules & Libraries
6. Typical Python Convention
7. How to write pythonic code

- **Data Structure and algorithms**

1. Arrays
2. Linked list
3. Binary trees, etc.
4. Time complexity

- **Basic Web Development**

1. Html language
2. Css language
3. Js language

- **API Development**

1. What is api ?
2. What is restapi is ?
3. What "http" is and various methods..

- In python framework
 1. Fastapi
 2. Flask
 3. django

- **Authentication & Authorization**

1. What are tokens ?
2. What is a JWT tokens ?
3. What is OAuth ?
4. How do you secure an api ?
5. Rate limiting

- **Pydantic** :- you just need to be familiar with APIs, building API and how you do that in python using the different frameworks

- **Database**

1. Inserting data
2. Retrieving data
3. Backing up data

- **Familiarize SQL Database**

1. How do you create tables ?
2. How do you insert values ?
3. How do you run advanced queries on the database ?

- **ORMs**

- **Linux & command line**

1. Navigating commands lines
2. Creating files
3. Removing files
4. Modifying configuration files

- **Git version control**

1. How to work with other people ?
2. What is a repository is?
3. How to make committed ?
4. How to change branch ?
5. Some advance features that are bound to come up(as a git user) ?

- **Testings**

1. Unit tests
2. Integration tests
3. System tests

- **Debugging**

1. How do you use your IDE ?
2. How do you step through the code and utilize a debugger ?
3. postman

- Docker how do you use this

- Kubernetes how do you use this

- **CI & CD**

1. How do you use github actions ?
2. How do you have automatic deployments run ?
3. How do you automatically test your code ?